

| Area | Title and Scoring Criteria                                                                                                                                                                                                                                                          | Points Possible |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1    | Mandatory Features                                                                                                                                                                                                                                                                  | 40              |
|      | Look up and display three additional aircraft information items other than route (e.g., Airline, Country, aircraft type)                                                                                                                                                            | 5               |
|      | Look up and display route (e.g., line or graphical display)                                                                                                                                                                                                                         | 5               |
|      | Look up and display aircraft point of origin and destination (e.g., JFK-PIT-LAX)                                                                                                                                                                                                    | 5               |
|      | Closest point of approach - Flight tracking of time, distance, altitude with nearby aircraft                                                                                                                                                                                        | 10              |
|      | Plot airports                                                                                                                                                                                                                                                                       | 5               |
|      | Filter and show tracks based upon minimum of two polygon or areas only                                                                                                                                                                                                              | 10              |
| 2    | Responsiveness / Performance<br>(Measured by selecting purge then monitor the desired action)                                                                                                                                                                                       | 30              |
|      | Loading tracks from Raspberry Pi ADS-B Station < 1 sec                                                                                                                                                                                                                              | 5               |
|      | Loading tracks from SBS < 1 sec                                                                                                                                                                                                                                                     | 5               |
|      | Looking up aircraft data for selected track < 1 sec                                                                                                                                                                                                                                 | 5               |
|      | Looking up / calculate CPA < 1 sec                                                                                                                                                                                                                                                  | 5               |
|      | Filter tracks based upon polygon < 1 sec                                                                                                                                                                                                                                            | 10              |
| 3    | Fault Detection and Error Handling<br>(Turn off wifi or remove USB connection)                                                                                                                                                                                                      | 30              |
|      | Identify loss communication with Raspberry Pi                                                                                                                                                                                                                                       | 5               |
|      | Recover from loss communication with Raspberry Pi                                                                                                                                                                                                                                   | 5               |
|      | Identify the loss of incoming RP ADS-B local tracks                                                                                                                                                                                                                                 | 5               |
|      | Recover from loss of incoming RP ADS-B local tracks                                                                                                                                                                                                                                 | 5               |
|      | Gracefully handle missing data to support features (e.g., display N/A when missing information, conducted by selecting purge then observe data elements fill in)<br>Hint - Use a playback file to show this capability                                                              | 10              |
| 4    | Architecture supports new requirements or enhancements<br>(Shown in presentation slides)                                                                                                                                                                                            | 30              |
|      | Demonstrate adding or removing features to tailor functionality: <b>5 points minimum, up to 10 points</b><br>(e.g., data analysis feature such as looking at all tracks heading west, how would you add this feature?, or how do you add more analysis to tracks within a polygon?) | 10              |
|      | Demonstrate that the remote user interface is capable of supporting the more computing intensive desired features: <b>5 points minimum, up to 10 points</b><br>(e.g., how does the additional data scale?)                                                                          | 10              |
|      | Demonstrate interfaces with other data sources: <b>5 points minimum, up to 10 points</b><br>(e.g., how does the new data interface with remote user interface?)                                                                                                                     | 10              |
| 5    | Map Support                                                                                                                                                                                                                                                                         | 20              |
|      | Show zooming in and out to an area outside of Pittsburgh                                                                                                                                                                                                                            | 8               |
|      | Show use of a different map data source (e.g., OpenStreet)                                                                                                                                                                                                                          | 8               |
|      | Discuss why you chose this map data option                                                                                                                                                                                                                                          | 4               |
| 6    | Remote User Interface                                                                                                                                                                                                                                                               | 40              |
|      | <b>Real time</b> display of maps and aircraft tracks (not just plotting tracks but projecting tracks when plots not received - show in the code)                                                                                                                                    | 5               |
|      | <b>Real time</b> display of "hooked" aircraft                                                                                                                                                                                                                                       | 5               |

|   |                                                                                                                                                                                                    |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|   | Display aircraft data and related metadata (e.g., show the difference between the original display and the new display)                                                                            | 5   |
|   | Implement click buttons (or similar feature) to remotely connect to the Raspberry Pi Flight Tracker, ADS-B Hub, recorded files, or other data files                                                | 5   |
|   | Implement slide bars or similar features to promote interaction of flight track data on the map visually (e.g., filter by altitude, speed, or aircraft type)                                       | 10  |
|   | Display polygons and show only aircraft that are within the defined waypoints on the map (e.g., square or irregular shape following an highway)<br>- Ease of use, user experience, display quality | 10  |
|   |                                                                                                                                                                                                    |     |
| 7 | Best user interface (UI)                                                                                                                                                                           | 10  |
|   | 1st Best UI: 10 points; 2nd Best UI: 8 points, 3rd Best UI: 6 points, 4th Best UI: 4 points, 5th Best UI: 4 points(points awarded based on client/sponsor judgment)                                | 10  |
|   |                                                                                                                                                                                                    |     |
|   |                                                                                                                                                                                                    |     |
|   | Total                                                                                                                                                                                              | 200 |

## Grading Scale

\*Note: Lines 2-4 show max points combinations for each task and the scaling based upon the evaluation (outstanding, satisfactory, marginal, unsatisfactory, not acceptable).

| Outstanding                                                                                                                                                                                                                                                          | Satisfactory                                                                                                                                                                                                          | Marginal                                                                             | Unsatisfactory                                                                                                  | Not Acceptable                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------|
| 10                                                                                                                                                                                                                                                                   | 8                                                                                                                                                                                                                     | 5                                                                                    | 3                                                                                                               | 0                             |
| 8                                                                                                                                                                                                                                                                    | 6                                                                                                                                                                                                                     | 4                                                                                    | 2                                                                                                               | 0                             |
| 5                                                                                                                                                                                                                                                                    | 4                                                                                                                                                                                                                     | 3                                                                                    | 1                                                                                                               | 0                             |
| 4                                                                                                                                                                                                                                                                    | 3                                                                                                                                                                                                                     | 2                                                                                    | 1                                                                                                               | 0                             |
| <p>Above and beyond</p> <p>Fully completed all aspects of the task. Demonstrated robust critical thinking through a comprehensive analysis that integrated relevant perspectives into an original, clear design rationale fully supported with consistent logic.</p> | <p>Completed all aspects of the task.</p> <p>Demonstrated critical thinking through a comprehensive analysis that integrated relevant perspectives into a clear design rationale supported with consistent logic.</p> | <p>No change to existing code base, but executes the functional task or feature.</p> | <p>Feature does not meet performance specifications.</p> <p>Feature does not meet the quality requirements.</p> | <p>Feature does not work.</p> |